

GEOTECHNICAL ENGINEERING - CV2040

Semester 4 (Jan-May) 2026

Instructors - Even Batch: RG Robinson (RGR), Ramesh Kannan Kandasami (RKK)

Instructors - Odd Batch: Tarun Naskar (TN) and Subhadeep Banerjee (SB)

Slot: C and T (Mon 10 Hrs, Tue 09 Hrs, Wed 08 Hrs, Fri 13 Hrs and Tutorial: Fri 14 to 16:50 Hrs)

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Place: offline (BSB 202 - Even Batch and BSB 203 - Odd Batch)

Course Objectives:

This course introduces the concepts of soil classifications, sub-soil investigation, mechanics of soils, analyses of geostructures, design of earth retaining structures, stability analysis of slopes and embankments, and design of foundations.

Expected Learning Outcomes:

At the end of the course, students will be aware of various theories, principles and techniques with respect to the fundamental behavior of soils and to design various geotechnical structures, such as retaining walls, embankments, dams, slopes, and foundations in varying soil conditions.

Lectures:

Offline: Please regularly attend the lectures, lab sessions, class tests and quizzes

Exam Schedule:

Quiz 1: 18th February 2026 - 08 to 08:50 Hrs

Quiz 2: 25th March 2026 - 08 to 08:50 Hrs

Final Exam: 08th May 2026 - 09 to 12 Hrs

Grading Policy:

Quiz 1: 15 marks

Quiz 2: 15 marks

Class test/ assignments: 20 marks (Class tests will be conducted after every module)

Final Exam: 50 marks (Entire syllabus)

Modules	Topics	Lab demonstration	Analytical exercise	Lectures
1 (RGR/TN)	<i>Introduction to geotechnical engineering:</i> Concepts of geology from a civil engineering perspective, formation geomaterials, application/importance, commonly encountered civil engineering materials, plate tectonics and earthquakes from structural geology perception, showcasing various minerals and their engineering performance.	<i>Laboratory-1:</i> Showcasing various geological samples, minerals, and their engineering performance		1-4
2 (RGR/TN)	<i>Soil classification:</i> Phase relationships. index properties, sieve & hydrometer analyses, soil classification.	<i>Laboratory-2:</i> Classification of soils (sieve & hydrometer analyses, Atterberg limits)		5-8
3 (RKK/TN)	<i>Stresses in soil:</i> Concepts of total stress, application of Boussinq, Westerguard & Newmark theories considering rectangular, trapezoidal, triangular, irregular loaded areas, concepts of effective stress, capillarity in soils, pore pressure in soils.		Problem-solving using excel and numerical tools such as Plaxis and Geostudio.	9-12
4 (RKK/TN)	<i>Permeability and seepage:</i> Darcy law, factors influencing the permeability, methods of determining permeability using laboratory and field techniques, seepage & flow nets.	<i>Laboratory-3:</i> Permeability of soil (falling head & constant head methods)	Development of flow nets using numerical tools such as Plaxis and Geostudio.	13-18

Modules	Topics	Lab demonstration	Analytical exercise	Lectures
5 (RGR/TN)	<i>Compressibility of soils:</i> Compaction and Consolidation.	<i>Laboratory-4:</i> Compaction and Consolidation experiment	Problem-solving using excel and numerical tools such as Plaxis and Geostudio.	19-26
6 (RGR/TN)	<i>Shear strength of soils:</i> State of stress in soil, Mohr circle, Mohr-Coulomb failure criteria, methods of assessing the soil's shear strength using direct shear, triaxial tests.	<i>Laboratory-5:</i> Direct shear, Unconfined compressive strength tests	Problem-solving using excel and numerical tools such as Plaxis and Geostudio.	27-34
7 (RKK/TN)	<i>Site investigation:</i> Methods of sampling, type of samples, in-situ tests such as SPT, CPT, vane shear test.	<i>Laboratory-6:</i> Field density measurement, Site visit for a demo of SPT, CPT tests.		35-38
8 (RKK/SB)	<i>Applied geotechnical engineering:</i> Rankine and Coulomb earth pressure theories, retaining structures Slope stability analyses ($\phi_u=0$ analysis, Ordinary method of slices) Bearing capacity theories, methods of improving bearing capacity, shallow foundation, pile foundations		Problem-solving using excel and numerical tools such as Plaxis and Geostudio.	39-54

References: The following are a restricted list of various interesting and useful books. Class notes and the books mentioned below will help you gain confidence in this course.

Text book:

- Ranjan, G. and Rao, A. S. R. (2022). *Basic and applied soil mechanics*, 4th Edition, New Age International Publishers, New Delhi, India.
- Knappett, J.A. and Craig, R.F. (2019). *Craig's Soil Mechanics*, 9th Edition, CRC Press, London, UK.

Reference books:

- Holtz, R. D., Kovacs, W. D., and Sheahan, T. C. (2010). *An Introduction to Geotechnical Engineering*, 2nd Edition, Pearson pubs.
- Das, B. M. (2010). *Principles of Geotechnical Engineering*, Cengage Learning, USA.
- Das, B. M. and Sivakugan, N. (2019). *Principles of Foundation Engineering*, (9th Edition), Cengage Learning, USA.
- Bowles, J. E. (2001). *Foundation Analysis and Design*, McGraw-Hill International, Singapore.
- Gulhati, S. K. and Datta, M. (2017). *Geotechnical Engineering*, Tata McGraw Hill, Delhi.
- Terzaghi, K., Peck, R. B., and Mesri, G. (1996). *Soil Mechanics in Engineering Practice (Third edition)*. John Wiley & Sons, Inc., New York.
- All Relevant Indian Standards (IS Codes) can be accessed from the Central Library, IIT Madras.

Academic Honesty: Lack of knowledge of the academic honesty policy is not a reasonable explanation for a violation.